

ADRENAL GLANDS

The adrenal glands are located on both sides of the body in retroperitoneum above and slightly medial to the kidneys. In humans, right adrenal gland is pyramidal in shape. But the left is semilunar or crescent shaped and somewhat larger. The adrenal glands measure approximately 3 cm in width, 5.0 cm in length, and up to 1.0 cm in thickness. Their combined weight in an adult human ranges from 7 to 10 grams. The glands are yellowish in colour.

The adrenal glands are surrounded by a fatty capsule and lie within the renal fascia. It also surrounds the kidneys. A weak septum of connective tissue separates the glands from the kidneys. The adrenal glands are directly below the diaphragm. They are attached to the crura of the diaphragm by renal fascia.

Each gland consists of two separate glandular tissues. The inner portion is the medulla. Its outer portion is the cortex. Cortex surrounds the medulla.

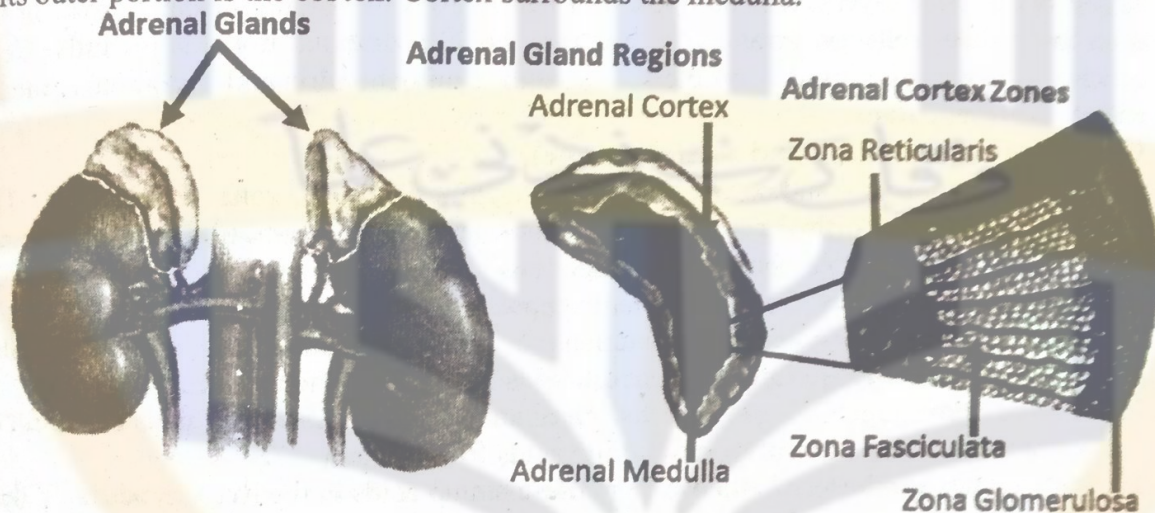


Fig: Adrenal Glands

Adrenal Cortex

The adrenal cortex is the outermost layer of adrenal gland. Within the cortex are three layers, called zones. Each layer has a distinct appearance, and each has a different function. The adrenal cortex produces hormones, aldosterone, cortisol, and androgens.

(a) Zona glomerulosa

The outermost zone of the adrenal cortex is zona glomerulosa. It lies immediately under the fibrous capsule of the gland. Cells in this layer form oval groups, separated by thin strands of connective tissue from fibrous capsule of gland. It has wide capillaries.

This layer is the main site for production of **aldosterone (mineralocorticoid)**. This hormone is produced by the action of enzyme **aldosterone synthase**.

Aldosterone (Mineralocorticoid)

The food intake or metabolic activity changes the amount of solutes in the blood. But aldosterone maintains concentrations of solutes like sodium in the extracellular fluid.

Aldosterone also promotes sodium and water reabsorption in the kidneys. Hence, it plays a major role in maintaining the homeostasis of extracellular fluid. Thus it plays an important role in regulation of blood pressure.

In the kidneys, aldosterone acts on the distal convoluted tubules and collecting ducts. It increases reabsorption of sodium and excretion of both potassium and hydrogen ions. Aldosterone is responsible for the reabsorption of about 2% of filtered glomerular filtration rates. Sodium retention is also a response of the distal colon and sweat glands to aldosterone receptor stimulation. Angiotensin II and extracellular potassium are two main regulators of aldosterone production. The amount of sodium present in the body affects the extracellular volume. It in turn influences blood pressure. Therefore, effects of aldosterone in sodium retention are important for the regulation of blood pressure.

(b) **Zona fasciculata**

Zona Fasciculata is situated between zona glomerulosa and zona reticularis. It is the largest of the three layers. It is composed of nearly 80% of volume of the cortex. In the zona fasciculata, cells are arranged in columns radially oriented towards medulla. Cells contain numerous lipid droplets, abundant mitochondria and complex smooth endoplasmic reticulum.

Glucocorticoids (cortisol and corticosterone)

Glucocorticoids cortisol and corticosterone are synthesized in zona fasciculata. The glucocorticoids like cortisol regulate metabolism and the concentration of blood sugar. They also defend the body in infections or tissue injury. Glucocorticoids and corticosterone causes immune system suppression.

Cortisol is the main glucocorticoid in humans. In species that do not produce cortisol, this role is played by corticosterone. Glucocorticoids have many effects on metabolism.

1. As their name suggests, they increase circulating level of glucose. This is the result of an increase in the mobilization of amino acids from protein.
2. It stimulates synthesis of glucose from these amino acids in the liver. In addition, they increase levels of free fatty acids. Cells can use these free fatty acids as an alternative to glucose to obtain energy.
3. Glucocorticoids also have effects unrelated to regulation of blood sugar levels. These are involved in suppression of immune system. These have a potent anti-inflammatory effect.
4. Cortisol reduces the capacity of osteoblasts to produce new bone tissue. Thus it decreases the absorption of calcium in the gastrointestinal tract.

Secretions

The adrenal gland secretes a basal level of cortisol. But it can also produce a great amount of hormone in response to **adrenocorticotrophic hormone (ACTH)** from anterior pituitary. Cortisol is not evenly released during the day. Its concentrations in the blood are highest in the early morning and lowest in the evening as a result of the circadian rhythm of ACTH secretion.

(c) Zona reticularis

The innermost cortical layer of cortex is zona reticularis. It lies directly adjacent to the medulla. Its small cells form irregular cords and clusters. These are separated by capillaries and connective tissue. The cells contain relatively small quantities of cytoplasm and lipid droplets. Sometimes display brown lipofuscin pigment. It produces androgens, mainly dehydroepiandrosterone (DHEA), DHEA sulfate (DHEA-S), and androstenedione (precursor to testosterone) in humans. These androgens are converted to fully functional sex hormones in gonads and other target organs. Adrenal cortex also secretes small amount of sex hormones. But these sex hormones have only a slight effect on male and female gonads. These sex hormones are composed of weak male hormones called **androgens**. They also have a small amount of female hormones **estrogens**.

Regulation of secretion of Adrenal Cortex Hormones

Glucocorticoids

Glucocorticoids are under the regulatory influence of hypothalamus-pituitary-adrenal (HPA) axis.

1. Glucocorticoid synthesis is stimulated by **adrenocorticotrophic hormone (ACTH)**. It is hormone released into the bloodstream by the anterior pituitary.
2. The production of ACTH itself is stimulated by the presence of **corticotropin-releasing hormone (CRH)**. It is released by neurons of the hypothalamus.
3. ACTH acts on the adrenal cells and it produces glucocorticoids.

The HPA axis is an example of a negative feedback system, in which cortisol itself acts as a direct inhibitor of both CRH and ACTH synthesis. The HPA axis also interacts with the immune system through increased secretion of ACTH at the presence of certain molecules of the inflammatory response.

Mineralocorticoid

Mineralocorticoid secretion is regulated mainly by **renin-angiotensin-aldosterone system (RAAS)**, concentration of potassium, and to a lesser extent concentration of ACTH.

1. Sensors of blood pressure in the juxtaglomerular apparatus of kidneys release enzyme **renin** into blood.
2. Renin starts a cascade of reactions. This reaction leads to formation of **angiotensin II**.
3. Angiotensin receptors in cells of zona glomerulosa recognize the substance. They stimulate the release of aldosterone.

Abnormalities of Adrenal gland

A number of endocrine diseases involve dysfunctions of adrenal gland.

1. Overproduction of cortisol leads cause Cushing's syndrome
2. The insufficient production is associated with Addison's disease.
3. Congenital adrenal hyperplasia is a genetic disease. It caused by dysregulation of endocrine control mechanisms.

4. A variety of tumors can arise from adrenal tissue. These are commonly found in medical imaging when searching for other diseases.

Adrenal Medulla

The adrenal medulla is under neural control. It contains neurosecretory cells. The adrenal medulla is present at the centre of each adrenal gland. It is surrounded by the adrenal cortex.

The adrenal medulla is driven by sympathetic nervous system via preganglionic fibers. These fibers originate in the thoracic spinal cord. It is innervated by preganglionic nerve fibers. Therefore, adrenal medulla can be considered as a specialized sympathetic ganglion. But adrenal medulla lacks distinct synapses and releases its secretions directly into the blood. The chromaffin cells of medulla are main source of catecholamines.

Catecholamines

Catecholamines include adrenaline and noradrenaline. These are water-soluble compounds. They have a structure made of a **catechol group** and an amine group. Adrenal glands produce most of adrenaline that circulates in the body. But it produces only for a small amount of circulating **noradrenaline**. Approximately 20% noradrenaline (norepinephrine) and 80% adrenaline (epinephrine) are secreted in adrenal medulla.

Fight-or-flight response

Adrenal medulla is activated during times of excitement, emergency, or stress. It prepared the body through sympathetic nervous system. It secretes epinephrine and norepinephrine. These hormones show **fight-or-flight response**. They prepared the body for emergencies. These hormones have following effects:

- They increase the heart rate.
- They increase the blood flow to vital organs.
- The airways in the lungs dilate. Therefore, more oxygen is delivered to all cells of the body.

Mechanism of Stress Response (fight-or-flight response)

Autonomic nervous system and HPA axis coordinate stress response to an acute threat. The rat detects cat and runs. Within seconds of threat, rat's sympathetic nervous system releases **catecholamines** (epinephrine and norepinephrine) from sympathetic nerve terminals and adrenal medulla. Its hypothalamic neurosecretory cells release CRH into hypothalamo-hypophyseal portal system.

A few seconds later, anterior pituitary secretes ACTH. Thus two output systems, sympathetic "fight-or-flight" system and HPA axis, together give response to a stressor. Their functions are not independent, but intermingled. For example, in addition to its role as a neurohormone that stimulates ACTH secretion, CRH also acts as a neurotransmitter in other areas of the brain. Here it stimulates the sympathetic nervous system.

Two Phases of Stress Response

The stress response includes two phases:

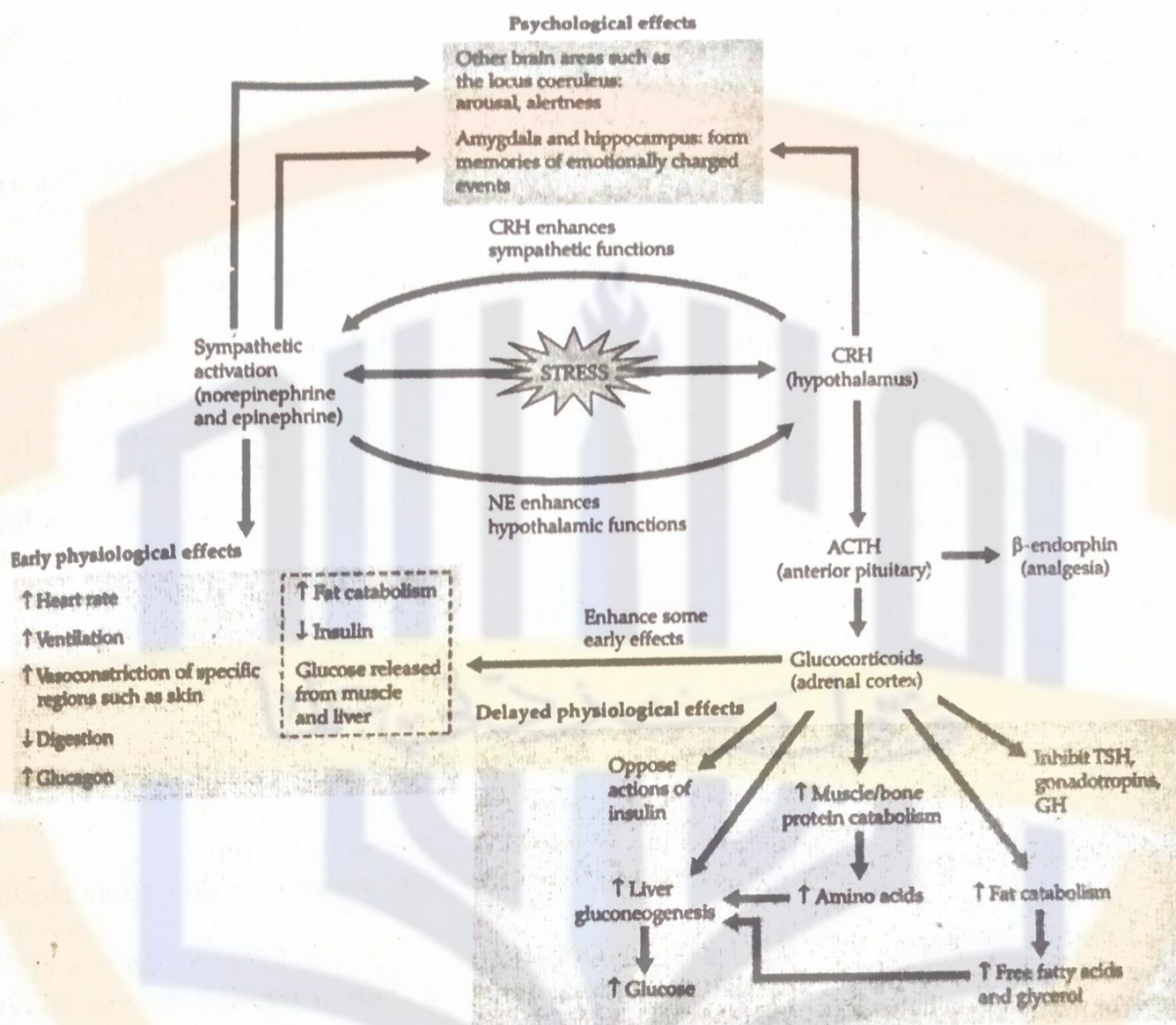


Fig: Mechanism of Stress Response

First Phase (Role of Catecholamines)

1. Within less than 1 min, catecholamines (epinephrine and norepinephrine) trigger increases in heart and respiration rates, blood pressure, and other sympathetic responses. These changes provide increased blood flow to skeletal muscles and heart.
2. These also cause increased air flow into and out of the lungs due to increase in diameter of bronchial airways.
3. Blood vessels to the skin constrict. It diverts the blood from sites of possible injury.
4. Digestive functions are suppressed.
5. Arousal of the CNS and alertness are promoted.
6. Epinephrine stimulates the release of glucose into the blood. It does it by triggering breakdown of glycogen stored in the liver and muscles. It also stimulates the release of fatty acids from lipid stores.

7. Epinephrine in the blood and norepinephrine from sympathetic nerve terminals both inhibit insulin secretion. They stimulate glucagon secretion from the islets of Langerhans. Thus catecholamines ensure high levels of glucose in blood to fuel physical exertion and maintain brain function.
8. Additional synergies occur during the first phase of the stress response. For example, epinephrine (in addition to CRH) stimulates the secretion of ACTH. The ACTH may do more than stimulate glucocorticoid secretion. ACTH is known to facilitate learning. Thus it may contribute to an animal's preparedness in responding to a similar stressor in the future.

Second Phase (glucocorticoids)

In the second phase of stress response, glucocorticoids are secreted by adrenal cortex.

1. Glucocorticoids reinforce the actions of the sympathetic nervous system. They facilitate the release of usable sources of energy into the bloodstream. They stimulate the catabolism of protein in muscle and (at high levels) bone. They stimulate the liver to use released amino acids to produce glucose in a process called **gluconeogenesis**. The liver cells release this newly formed glucose into the blood.
2. Like epinephrine and norepinephrine, glucocorticoids oppose the action of insulin and ensure fuel availability.
3. Glucocorticoids also stimulate catabolism of fats. Hence fatty acids can be used as an alternative energy source by all tissues except the brain.
4. The metabolic actions of glucocorticoids, coordinated with catecholamines, ensure glucose availability to the brain for physical exertion and possible fasting. The amino acids released by protein catabolism are also available for tissue repair.
5. Glucocorticoids increase their permissive effect on vasoconstriction stimulated by catecholamines.
6. They also inhibit the secretion of gonadotropins (FSH and LH), thyrotropin (TSH) and growth hormone (GH) from anterior pituitary. Thus it inhibits the reproductive and growth functions.

Safe in its burrow, the rat experiences diminished sympathetic neural responses. The glucocorticoid molecules in its general circulation feedbacks negatively on CRH and ACTH cells of HPA axis. Thus the glucocorticoids themselves modulate the stress response. With decreased ACTH in the circulation, the adrenal cortex secretes decreased amounts of glucocorticoids. Thus its concentrations in the blood return to basal levels.

OVARY

Hormones of Ovaries

Ovaries produce and release two groups of sex hormones, progesterone and estrogen. These substances work together to promote healthy development of female sex characteristics during puberty and to ensure fertility.

There are four major classes of ovarian hormones. These hormones regulate the female reproductive functions.

1. **Estrogens (estrin, estrone, and estradiol):** It regulates the menstrual and estrus

cycles. It stimulates the development of the mammary glands and other female secondary sexual characteristics.

2. **Progestins (primarily progesterone):** It regulates the menstrual and estrus cycles. It also regulates the development of the mammary glands. It helps in placenta formation during pregnancy.
3. **Relaxin:** It is produced in small quantities. It softens the opening of the uterus (cervix) at the time of delivery.
4. **Inhibin:** It inhibits the secretion of FSH.

Estrogens

Estrogen is primary female sex hormone. It is responsible for development and regulation of the female reproductive system and secondary sex characteristics. There are three major endogenous estrogens in females. All these have estrogenic hormonal activity. These are estrone (E1), estradiol (E2), and estriol (E3). The estrane steroid estradiol is the most potent and prevalent of these. Another type of estrogen called estetrol (E4) is produced only during pregnancy. Estrogens circulate at lower levels than androgens in both men and women. While estrogen levels are significantly lower in males compared to females. Estrogens also have important physiological roles in males.

Types of Estrogens

There are four major naturally occurring estrogens in women. These are estrone (E1), estradiol (E2), estriol (E3), and estetrol(E4).

1. **Estradiol (E2):** It is the predominant estrogen during reproductive years.
2. **Estrone (E1):** During menopause, estrone is the predominant circulating estrogen.
3. **Estriol (E3):** During pregnancy estriol is the predominant circulating estrogen.
4. **Estetrol (E4):** It is another type of estrogen. It is produced only during pregnancy.

Estradiol is the most important estrogen in non-pregnant females. But during pregnancy this role shifts to estriol. In postmenopausal women estrone becomes primary form of estrogen in the body.

Mode of Action of Estrogens

The actions of estrogen are mediated by **estrogen receptor (ER)**. It is a dimeric nuclear protein. It binds to DNA and controls gene expression. Like other steroid hormones, estrogen enters passively into cell. Here it binds to and activates estrogen receptor. The **estrogen-ER complex** binds to specific DNA sequences called a **hormone response element** to activate transcription of target genes. ER is expressed in specific tissues including the ovary, uterus and breast.

Functions of Estrogens

Estrogens are present in both men and women. They are usually present at significantly higher levels in women of reproductive age. They promote the development of female secondary sexual characteristics like breasts. They are also involved in the thickening of the endometrium and other aspects of regulating menstrual cycle. In males, estrogen regulates certain functions of reproductive system important to the maturation of sperm. It may be necessary for a healthy sexual desire.

Progesterone

As the egg migrates down the fallopian tube, progesterone is released. It is secreted by a temporary gland formed within ovary after ovulation called **corpus luteum**. Progesterone prepares the body for pregnancy by thickening the uterine lining. If a woman is not pregnant, the corpus luteum disappears and menstruation occurs.

Progesterone has a number of physiological effects. These effects are amplified in the presence of estrogens. Progesterone is sometimes called hormone of pregnancy. It has many roles relating to the development of the fetus:

1. Progesterone converts endometrium to its secretory stage to prepare the uterus for implantation. At the same time progesterone affects the vaginal epithelium and cervical mucus. It makes epithelium thick and impenetrable to sperm.
2. Progesterone is anti-mitogenic in endometrial epithelial cells. Normal menstrual bleeding is progesterone-withdrawal bleeding. If ovulation does not occur and the corpus luteum does not develop, levels of progesterone may be low. It leads to an ovulatory dysfunctional uterine bleeding.
3. During implantation and gestation, progesterone decrease maternal immune response to allow for acceptance of the pregnancy.
4. Progesterone decreases contractility of the uterine smooth muscle.
5. In addition progesterone inhibits lactation during pregnancy. The fall in progesterone levels following delivery is one of the triggers for milk production.
6. A drop in progesterone levels is possibly one step that facilitates the onset of labor. The fetus metabolizes placental progesterone in the production of adrenal steroids.

TESTIS

Testes are male gonads located in scrotal sacs. The testis secretes steroid and protein hormones. Androgens are main steroids produced by testis. **Inhibins** and **activins** are proteins secreted by testes. **Anti-Mullerian hormone (AMH)** or **Mullerian-inhibiting substance (MIS)** is another protein of testicular origin. All these hormones play an important role in testicular development and maturation.

Testosterone

Testosterone has following functions:

1. It acts with luteinizing and follicle-stimulating hormones and stimulate spermatogenesis.
2. Testosterone is necessary for the growth and maintenance of the male sex organs
3. It promotes the development and maintenance of sexual behavior.
4. It stimulates the growth of facial and pubic hair in humans. It causes the enlargement of the larynx. Enlarged larynx deepens the voice.
5. It promotes increase in muscle mass.
6. It increases body height.

Control of Testosterone Production

The hypothalamus and pituitary gland control the production of testosterone. The hypothalamus sends a signal to pituitary gland to release gonadotrophic substances

(follicle stimulating hormone and luteinizing hormone). Luteinizing hormone (LH) stimulates testosterone production. If too much testosterone is produced, the hypothalamus alerts the pituitary gland to make less LH. It decreases the testosterone production.

PLACENTA

In addition to its role in transporting molecules between mother and fetus, the placenta is a major endocrine organ. Placenta synthesizes a huge and diverse number of hormones and cytokines. All these have major influences on ovarian, uterine, mammary and fetal physiology. Placenta produces some major steroid and protein hormones.

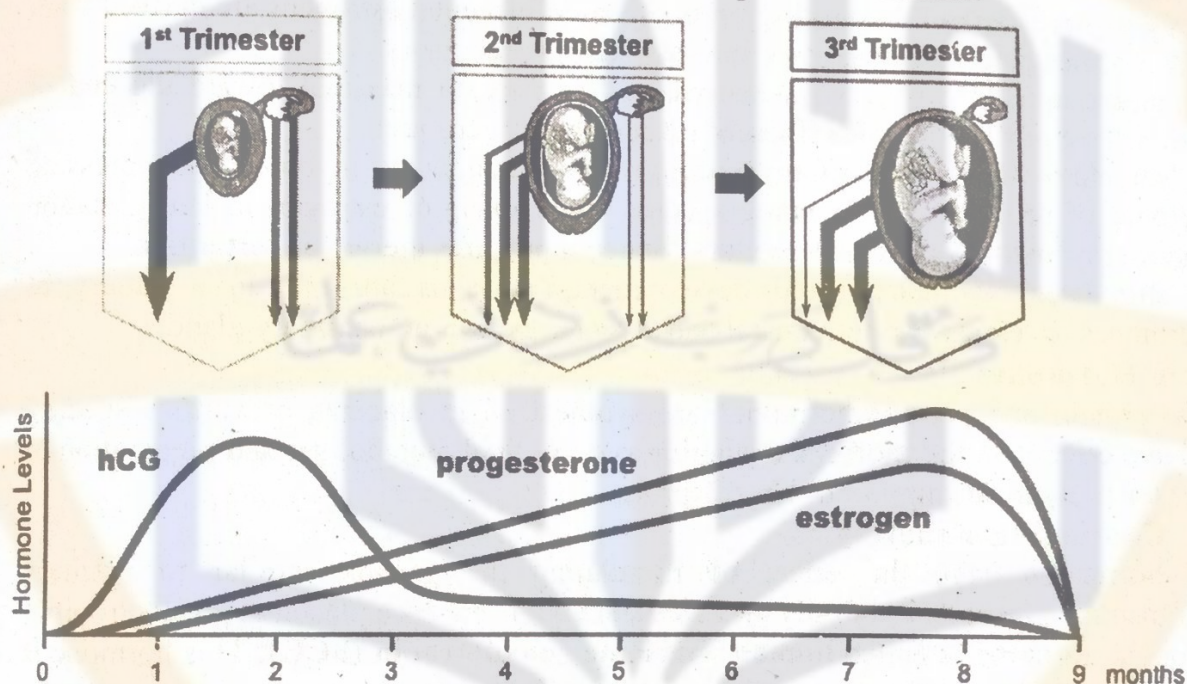


Fig: Hormones of placenta

Steroid Hormones

Sex steroids are the best known examples of placental hormones. Two major groups are produced by all mammals:

1. Progestins

Progestins are molecules that bind to the progesterone receptor. Progesterone itself is called hormone of pregnancy. It plays a critical role in supporting endometrium and hence on survival of conceptus.

The placenta of all mammals produces progestins. Its quantity varies significantly. In some species (women, horses, sheep, cats), sufficient progestin is secreted by placenta. Hence ovaries or corpora lutea can be removed after formation of the placenta and the pregnancy will continue. In other animals (cattle, pigs, goats, dogs), luteal progesterone is necessary throughout gestation. Their placenta does not produce sufficient amounts.

Progestins, including progesterone, have two major roles during pregnancy:

- a) Support of endometrium to provide an environment helpful for fetal survival. If the endometrium is not given progestins, the pregnancy will be terminated.
- b) Suppression of contractility in uterine smooth muscle. This is called **progesterone block** on myometrium. Toward the end of gestation, this myometrial-blocking effect is stopped by rising levels of estrogens. It starts parturition.
- c) Progesterone and other progestins also inhibit secretion of the pituitary gonadotropins luteinizing hormone and follicle stimulating hormone. This effect almost always prevents ovulation from occurring during pregnancy.

2. Estrogens

The placenta produces several distinct estrogens. In women, the major estrogen produced by the placenta is **estriol**. Depending on the species, placental estrogens are derived from fetal androgens, placental progestins, or other steroid precursors.

The concentration of estrogens in maternal blood rises to maximal toward the end of gestation. Two of the principle effects of placental estrogens are:

- a) It stimulates the growth of myometrium and antagonizes myometrial-suppressing activity of progesterone. In many species, high levels of estrogen in late gestation induce myometrial oxytocin receptors. Hence it prepares uterus for parturition.
- b) It stimulates mammary gland development. Estrogens are one in a battery of hormones necessary for both ductal and alveolar growth in mammary gland.

Protein Hormones

Several protein and peptide hormones are synthesized in placenta of various species. They have effects on the mother's endocrine system, fetal metabolism and preparation of mother for postpartum support of her offspring.

1. Chorionic gonadotropins

These hormones have the effect of stimulating the gonads, similar to pituitary gonadotropins. The species of only primates and equids produce placental gonadotropin.

The human hormone is called **human chorionic gonadotropin (hCG)**. This hormone is produced by fetal trophoblast cells. It binds to the luteinizing hormone receptor on cells of the corpus luteum. Thus it prevents luteal regression. Thus, hCG serves as signal for maternal recognition of pregnancy.

Equine chorionic gonadotropin is also produced by fetal trophoblast cells. It is actually the same molecule as equine luteinizing hormone.

2. Placental lactogens

These hormones are molecular relatives of prolactin and growth hormone. These hormones are identified in primates, ruminants and rodents, but not in other species.

The functions of placental lactogens are not well understood. They modulate fetal and maternal metabolism. They mobilize energy substrates for fetal use. In some species they stimulate function of corpus luteum. They participate in development of the mammary gland prior to parturition.

3. **Relaxin**

Relaxin act synergistically with progesterone to maintain pregnancy. It also causes relaxation of pelvic ligaments at end of gestation. Therefore, it helps in parturition. In some species in which relaxin is produced by the placenta. But in others, the major source is corpus luteum. In some species, relaxin is produced by both the corpus luteum and placenta.

PANCREATIC ISLETS

Irregularly shaped patches of endocrine tissue located within the pancreas of most vertebrates. They are named for German physician Paul Langerhans, who first described them in 1869. The normal human pancreas contains about 1,000,000 islets. The islets consist of four distinct cell types. Three types of cells (alpha, beta, and delta cells) produce important hormones. But the fourth component (C cells) has no known function.

- a) Alpha cells producing glucagon (20% of total islet cells)
- b) Beta cells producing insulin and amylin ($\approx 70\%$)
- c) Delta cells producing somatostatin ($<10\%$)
- d) Epsilon cells producing ghrelin ($<1\%$)
- e) PP cells (gamma cells or F cells) producing pancreatic polypeptide ($<5\%$)

Islets can influence each other through paracrine and autocrine communication

The paracrine feedback system of the pancreatic islets has the following structure:

- a) Glucose/Insulin: activates beta cells and inhibits alpha cells
- b) Glycogen/Glucagon: activates alpha cells which activates beta cells and delta cells
- c) Somatostatin: inhibits alpha cells and beta cells.

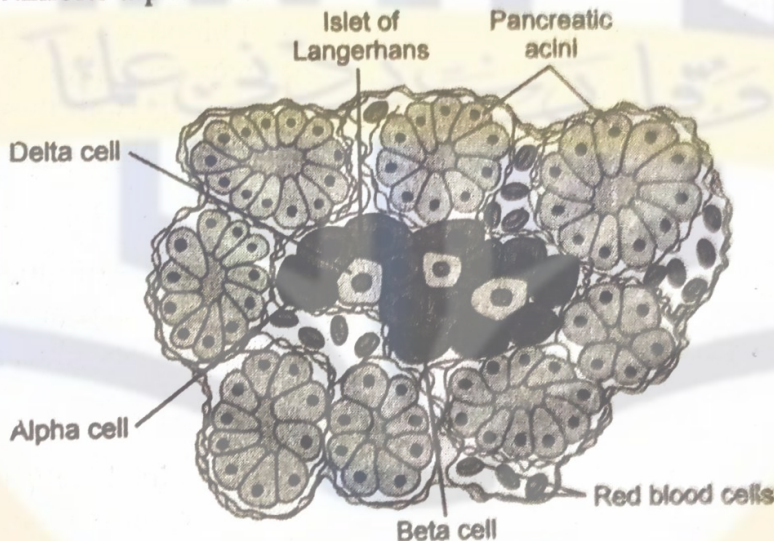


Fig: Inner structure of Pancreatic Islets

1. Beta Cells

The most common islet cells are beta cells. They produce insulin. It is a major hormone in the regulation of carbohydrate, fat, and protein metabolism. Insulin is crucial in several metabolic processes:

- 1. Promotes the uptake and metabolism of glucose by the body's cells.
- 2. It prevents release of glucose by the liver

3. It causes muscle cells to take up amino acids. Amino acids are basic components of protein

4. It inhibits the breakdown and release of fats.

The release of insulin from the beta cells can be triggered by growth hormone (somatotropin) or by glucagon. But the most important stimulator of insulin release is glucose. When the blood glucose level increases after a meal then insulin is released to counter it. The inability of the islet cells to make insulin or the failure to produce sufficient amount to control blood glucose level are causes of diabetes mellitus.

2. Alpha Cells

The alpha cells of the islets of Langerhans produce an opposing hormone, glucagon. It releases glucose from the liver and fatty acids from fat tissue. In turn, glucose and free fatty acids favour insulin release and inhibit glucagon release.

3. Delta Cells

The delta cells produce somatostatin. It is a strong inhibitor of somatotropin, insulin, and glucagon. Its role in metabolic regulation is not yet clear. Somatostatin is also produced by hypothalamus. Its function is to inhibit secretion of growth hormone pituitary gland.

4. F Cells

A fourth type of islet cell, F (or PP) cell, is located at the periphery of islets and secretes pancreatic polypeptide. These hormones regulate one another's secretion through paracrine cell-cell interactions.

Blood Sugar Regulation

Blood sugar regulation is the process by which levels of blood sugar, primarily glucose, are maintained by body within a narrow range. This tight regulation is called glucose homeostasis. Insulin lowers blood sugar and glucagon raises it. Both these hormones are the most well known of hormones involved. But some other glucoregulatory hormones also play role in it.

Mechanism

Blood sugar levels are regulated by negative feedback. The levels of glucose in the blood are monitored by many tissues. But the cells in the pancreatic islets are among the most well understood for regulating blood sugar level. Two hormones glucagon and insulin play role in it:

(a) Glucagon

If the blood glucose level falls to dangerous levels the alpha cells of the pancreas release glucagon. It is a hormone. It effects on liver cells to increase blood glucose levels. They convert glycogen into glucose by glycogenolysis. The glucose is released into the bloodstream. It increases the blood sugar. Hypoglycemia is the state of low blood sugar. It is treated by restoring the blood glucose level to normal by the ingestion or administration of dextrose or carbohydrate foods. It is often self-diagnosed and self-medicated orally by the ingestion of balanced meals. In more severe circumstances, it is treated by injection or infusion of glucagon.

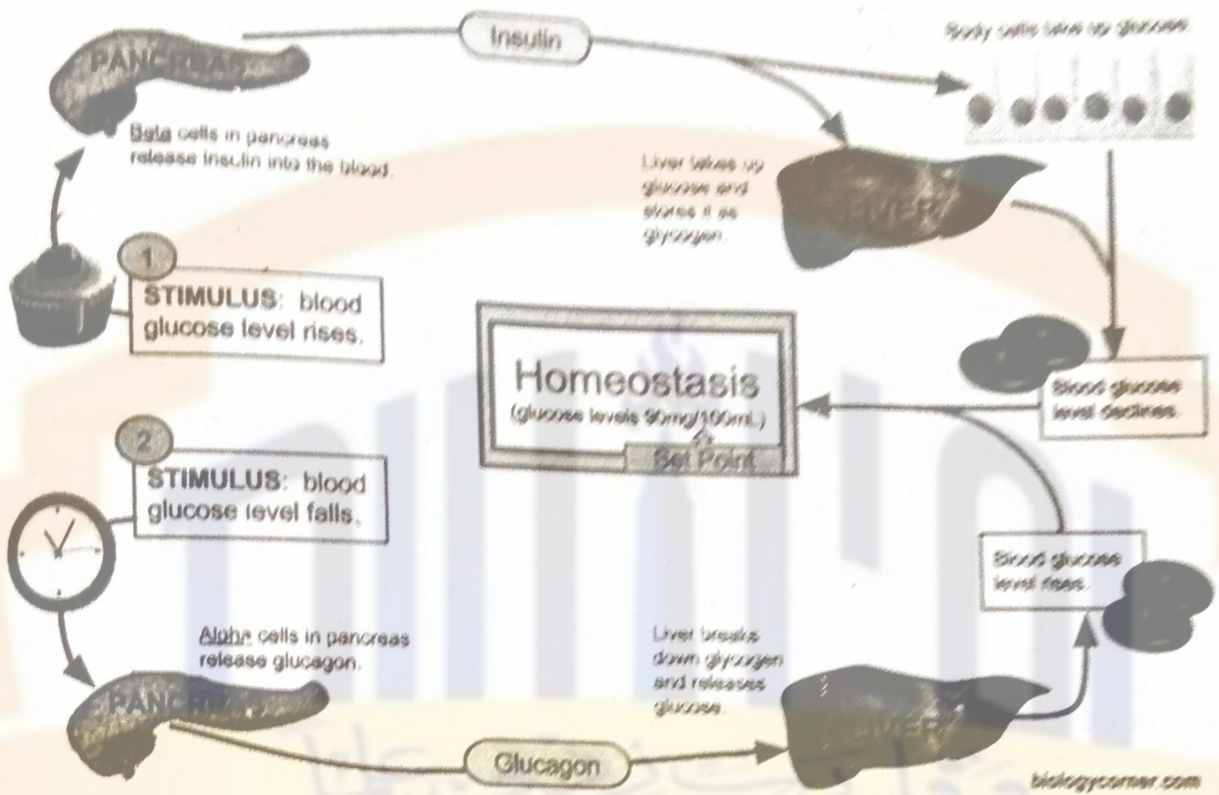


Fig: Glucose Homeostasis

(b) Insulin

Sometimes levels of blood sugar rise. It may rise as a result of glycogen conversion, or from digestion of a meal. As a result, different insulin is released from beta cells found in the Islets of Langerhans in pancreas. Insulin activates the liver to convert more glucose into glycogen. This process is called glycogenesis. Insulin forces about 2/3 of body cells (primarily muscle and fat tissue cells) to take up glucose from the blood through the GLUT4 transporter. Thus it decreases blood sugar. When insulin binds to the receptors on cell surface, vesicles containing GLUT4 transporters come to the plasma membrane. They fuse together by the process of endocytosis. It enables a facilitated diffusion of glucose into the cell. The glucose enters the cell. It is phosphorylated into Glucose-6-Phosphate in order to preserve the concentration gradient. Thus glucose will continue to enter the cell. Insulin also provides signals to several other body systems. It is the chief regulator of metabolic control in humans.

Diabetes Mellitus

Diabetes mellitus is a syndrome of impaired carbohydrate, fat, and protein metabolism. It is caused by lack of insulin secretion or decreased sensitivity of the tissues to insulin. There are two general types of diabetes mellitus:

1. **Type I diabetes:** It is also called **insulin-dependent diabetes mellitus (IDDM)**. It is caused by lack of insulin secretion.

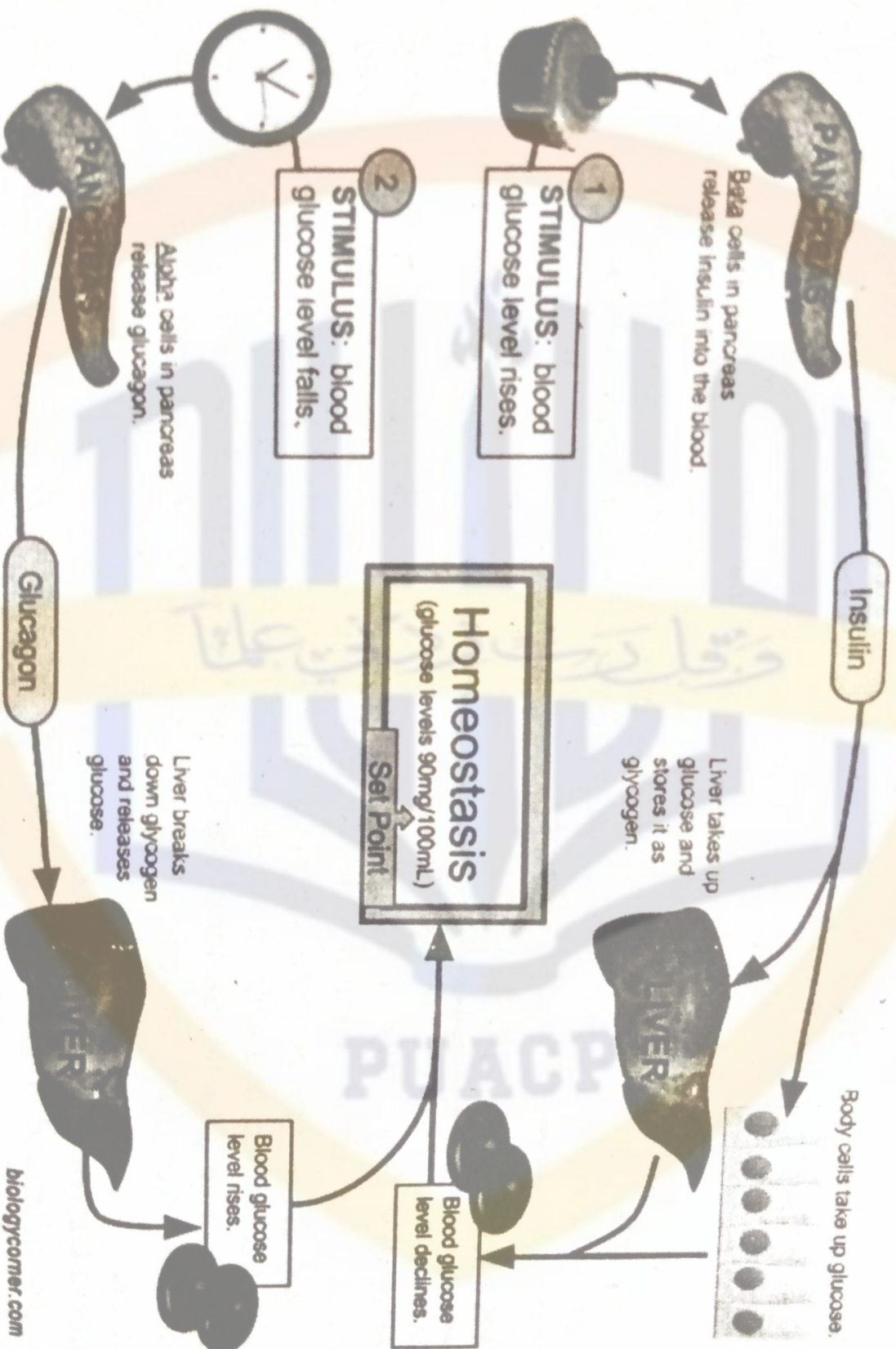


Fig: Glucose Homeostasis

2. Type II diabetes: It is also called **non-insulin-dependent diabetes mellitus (NIDDM)**. It is caused by decreased sensitivity of target tissues to the metabolic effect of insulin. This reduced sensitivity to insulin is called **insulin resistance**.

In both types of diabetes mellitus, metabolism of all the main foodstuffs is altered. The basic effect of insulin lack or insulin resistance is on glucose metabolism. It prevents the efficient uptake and utilization of glucose by most cells of the body. As a result, blood glucose concentration increases. Cell utilization of glucose falls increasingly lower. The utilization of fats and proteins